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Independent control of voluntary movements and associated anticipatory postural responses in a bimanual task

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Abstract

Objective: When one hand loads the other arm, EMG responses in the stationary arm anticipate the load. This study used transcranial magnetic stimulation over each hemisphere to clarify the relationship between a voluntary movement on one side and the anticipatory postural response on the other.

Methods: Subjects ($n=7$) performed elbow flexion movements of one arm as a reaction-time task. Because subjects' arms were linked, flexion about one elbow resulted in extension force about the other, and an anticipatory response occurred in those elbow flexor muscles. After the 'go' signal and before the predicted onset of EMG, transcranial magnetic stimuli were delivered over one or other motor cortex.

Results: Stimulation contralateral to the reaction-time movement delayed the onset of voluntary EMG (46 ms in right biceps, 77 ms left) but did not alter the onset of EMG in the postural arm. Stimulation contralateral to the anticipatory postural response delayed only the postural EMG (left 96 ms, right 52 ms).

Conclusions: Thus, the associated voluntary and postural responses were delayed independently by stimuli over their respective contralateral motor cortex.

Significance: This suggests that, although timing of responses may be linked by an initial signal, the response from each motor cortex develops independently.

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Keywords: Transcranial magnetic stimulation; Reaction time; Motor control; Anticipatory response; Postural response

1. Introduction

When one hand is used to load or unload the other arm, EMG responses occur in the stationary arm in anticipation of the change of load. The responses in the postural limb are linked in time with the movements of the other limb and scale with the load change (e.g. Bennis et al., 1996; Hugon et al., 1982; Lacquaniti and Maioli, 1989). They occur before any possible reflex response to changes in afferent firing. Similar responses for interactions between leg and arm can be learnt in three or four trials (Forget and Lamarre, 1995). As long as a voluntary movement is responsible for the change of load, an anticipatory postural response can be acquired although more practice trials are required if the link is remote (e.g. pressing a button to unload the other arm; Diedrichsen et al., 2003). Thus, anticipatory postural responses appear to be learned responses of which subjects are unaware and which become part of the motor program which controls the voluntary movement.

It has been suggested that when an intended voluntary movement is made with one side of the body and an anticipatory postural response occurs in muscles on the other side of the body, both may be controlled by the same cerebral hemisphere (Viallet et al., 1992). This could possibly occur at the level of the motor cortex, with contralateral descending control of the voluntary movement and ipsilateral control of the postural response. Evidence against this is provided by patients with corticospinal lesions who lose anticipatory responses in the contralateral arm during an unloading task (Viallet et al., 1992). However, in a loading task, hemiparetic patients with

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cortical or subcortical lesions did make anticipatory postural responses, although these were slower and longer than normal (Bennis et al., 1996).

Transcranial magnetic stimulation over the motor cortex causes an initial excitatory response which can be seen as a motor evoked potential (MEP) in the electromyogram (EMG). During a voluntary contraction a long period of EMG silence of up to 200–250 ms follows the MEP. While inhibition of the motoneurones contributes to the initial part of this silent period, the latter part (>100 ms) is thought to be due to intracortical inhibition and interruption of cortical outflow (Chen et al., 1999; Fuhr et al., 1991; Inghilleri et al., 1993). Intracortical inhibition probably comes about through the activation of inhibitory interneurones by the transcranial magnetic stimulus and may involve the actions of GABA (Roick et al., 1993; Siebner et al., 1998). When a magnetic stimulus is delivered just before a voluntary movement, the movement is delayed. The onset of EMG occurs later than expected but the pattern of EMG activation is maintained. For example, for a ballistic movement the triphasic agonist-antagonist-agonist EMG bursts still occur and the movement itself is successfully performed (Day et al., 1989; Pascual Leone et al., 1992; Taylor et al., 1995; Ziemann et al., 1997). This delay of a voluntary movement seems to be related to the silent period in an ongoing contraction. That is, to an inhibition of output from the motor cortex and not to effects from stimulation of other motor areas (Taylor et al., 1995; Ziemann et al., 1997).

In the current study, subjects performed reaction time movements in which a voluntary flexion of one arm was accompanied by an anticipatory postural response in the other arm. Transcranial magnetic stimuli were applied over the motor cortex of each hemisphere to attempt to delay the movement. If the prime mover and the postural response were controlled from the one motor cortex, stimulation over one hemisphere should delay both actions. The study finds that this does not occur. Instead, the prime mover action and the postural response are delayed independently by stimulation over the motor cortex contralateral to each of the contractions.

2. Methods

Eleven normal subjects (21–43 years; 5 male) took part in experiments to investigate the effect of transcranial magnetic stimulation on tasks which involved voluntary flexion of one elbow and a linked postural response of the flexors of the other elbow. Experiments were approved by the institutional ethics committee and all subjects gave informed consent to the study.

Subjects sat with both elbows resting on a table in front of them. Forearms were held close to vertical and supinated. A length of rope was fastened to the wrists via wide bands. The rope led from the right wrist directly anterior to a pulley. It then passed to another pulley anterior to the left arm and back to the left wrist (see Fig. 1). A stiff spring in series with the rope formed part of the linkage. With this set up, flexion of one elbow pulled the other elbow into extension but this pull could be resisted by stretching the spring. The spring had a stiffness of ~0.75 N/mm which was doubled for stronger (male) subjects. A force transducer (Xtran, Melbourne, Australia) also in series with the rope recorded tension between the two wrist. EMG was recorded from both right and left biceps brachii through Ag/AgCl surface electrodes placed over each muscle in a belly-tendon configuration. EMG was filtered (10 Hz-1 kHz) and amplified and recorded to disk along with tension. All signals were sampled at 1 kHz (CED 1401 using Signal software). Transcranial magnetic stimulation was delivered through a figure-8 coil (9 cm; Magstim 200, Magstim Co,Wales, UK) over the optimal site to evoke responses in the contralateral biceps brachii. Stimulation was sometimes over the right hemisphere and at other times over the left hemisphere. Each site was marked on the scalp to keep the coil positions constant. In each subject, stimuli of the same intensity (80% of stimulator output) were delivered over each hemisphere.

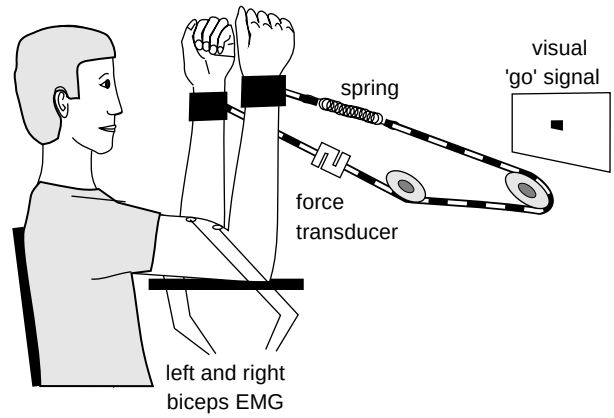

Fig. 1. Experimental set-up. When the subject flexed one elbow, the other arm was pulled into extension via the rope which passed around fixed pulleys. The spring enabled arm position to be maintained if the pull was resisted.

Eight subjects performed 7 sets of 20 trials on each of two days. In each trial, subjects were required to make a reaction time response. A red warning light (an LED) was followed 0.8–1.2 s later by a green 'go' signal. Subjects were instructed to flex one elbow as quickly as possible on the 'go' signal and keep the other arm still. Trials were performed about 10 s apart. In one experimental session, subjects flexed the left elbow and in the other session, the right elbow. The left-right order was randomised between subjects. In some trials, subjects received a transcranial magnetic stimulus in the interval between the 'go' signal and the onset of EMG in the muscle performing the voluntary elbow flexion.

In each session, subjects performed practice trials until comfortable with the task. An initial set of 20 trials was then collected. On the basis of these trials, the transcranial

magnetic stimulus was timed to occur approximately 40 ms before the onset of EMG. Timing varied from 80 to 120 ms after the 'go' signal in different subjects, but was the same for all conditions in each individual. Subjects then performed 4 sets of 20 trials. In each of these sets, transcranial magnetic stimuli were delivered in half the trials in a pseudo-random fashion. In 2 of the 4 sets, the magnetic stimuli were delivered over the left hemisphere and in the other 2 sets, over the right hemisphere. Subjects then completed a further set of trials in which no transcranial magnetic stimuli were given. Finally, subjects sat with the eyes closed while unexpected perturbations were delivered to the arm which had not performed the voluntary flexion task during the session. These perturbations pulled the arm into extension and were delivered by the experimenter pulling the rope attached to the wrist. Because the rope continued to run through the pulley, pulls were in the same direction as those delivered to the arm by the subjects' own voluntary flexions of the other elbow. Over the first 100 ms of the unexpected pull, force changed by an average of 88% (range 50–140%) of the change seen during the voluntary task. Extension of the perturbed arm over the same time was 1.6–6 times greater with the unexpected pull than with the voluntary task. Thus, the unexpected pull was probably the more effective stimulus in evoking stretch reflexes in the elbow flexor muscles.

Thus, each subject completed 120 trials in which the voluntary task was to flex the right elbow. Of these trials, 40 were collected in sets in which no stimuli were delivered; 20 were affected by stimulation over the left hemisphere and were collected mixed with 20 trials without stimulation; and 20 were affected by stimulation over the right hemisphere and were collected mixed with 20 trials without stimulation. Each subject also completed 120 trials in which the voluntary task was to flex the left elbow. Finally, 20 responses to unexpected perturbations of each arm were collected.

The study was repeated in 5 subjects of whom 2 had taken part in the first experiment. In this second experiment, the intensity of transcranial magnetic stimulation was set for each hemisphere of each subject to evoke silent periods of about 180 ms in the contralateral biceps brachii during a weak voluntary contraction. Stimulus intensities ranged from 85 to 95% stimulator output.

2.1. Data analysis

The onset of EMG in the voluntary and postural responses was measured for each trial for each subject. The latency of EMG onset was taken as the time from the 'go' signal to the start of continuous EMG and was measured by cursor. Timing of the perturbation to the arm which subjects held still was also measured in individual trials. It was measured as the start of increased tension in the link between the arms, or as the start of extension force when the pull was applied unexpectedly. Statistical analysis include unpaired *t*-tests with Bonferroni correction (significance set at $p<0.003$), repeated measures analysis of variance with post hoc comparison between the means (Student–Newman–Keuls; significance set at $p<0.05$). Data are expressed as mean $\pm$ SD.

3. Results

Subjects performed reaction time tasks in which a voluntary flexion of one arm was accompanied by an anticipatory postural response in biceps brachii of the other arm. Transcranial magnetic stimulation, delivered over one hemisphere, delayed the onset of EMG in the contralateral biceps brachii but not in the ipsilateral muscle. That is, magnetic stimulation of the motor cortex delayed the voluntary movement and its associated postural response independently.

When subjects performed the voluntary task of flexing one arm, and in doing so pulled the other arm into extension, an EMG response was evoked in biceps brachii in the arm that subjects attempted to hold still (see Fig. 2A). On average, this response occurred 15 ± 20 ms after the time at which tension increased in the linkage between the wrists. In contrast, when a pull into extension was unexpectedly applied to the arm, the EMG response in biceps brachii started after 97 ± 11 ms (mean $\pm$ SD). For each arm in each subject, the time from the increase in force to the start of EMG evoked by the unexpected pull was compared to that occurring during the voluntary task. In 7 of 8 subjects, the time from force onset to the start of EMG during the voluntary task (range -5–32 ms) was significantly shorter than after the unexpected pull (Fig. 2B; range 77–119 ms; *t*-tests with Bonferroni correction). Thus, EMG in the arm which was held still during the reaction time task did not occur in response to the increase in force and consequent extension of the elbow, initiated by the voluntary flexion of the other arm, but anticipated this increase. Data from the one subject in whom this anticipatory response did not occur was excluded from further analysis.

3.1. Control reaction times

In some sets of trials no transcranial magnetic stimuli were given. When subjects ($n=7$) flexed the right arm in response to the 'go' signal while holding the left arm still, the onset of EMG in the right arm (prime mover) was at 146 ± 16 ms after the 'go' and that in the left arm (postural response) was at 183 ± 20 ms. When subjects flexed the left arm while holding the right arm still, the onset of EMG in the prime mover (left arm) was at 150 ± 18 ms and in the postural arm (right arm) was at 191 ± 31 ms.

3.2. Effect of transcranial magnetic stimulation

Transcranial magnetic stimulation over one hemisphere delayed the onset of EMG in the contralateral biceps brachii

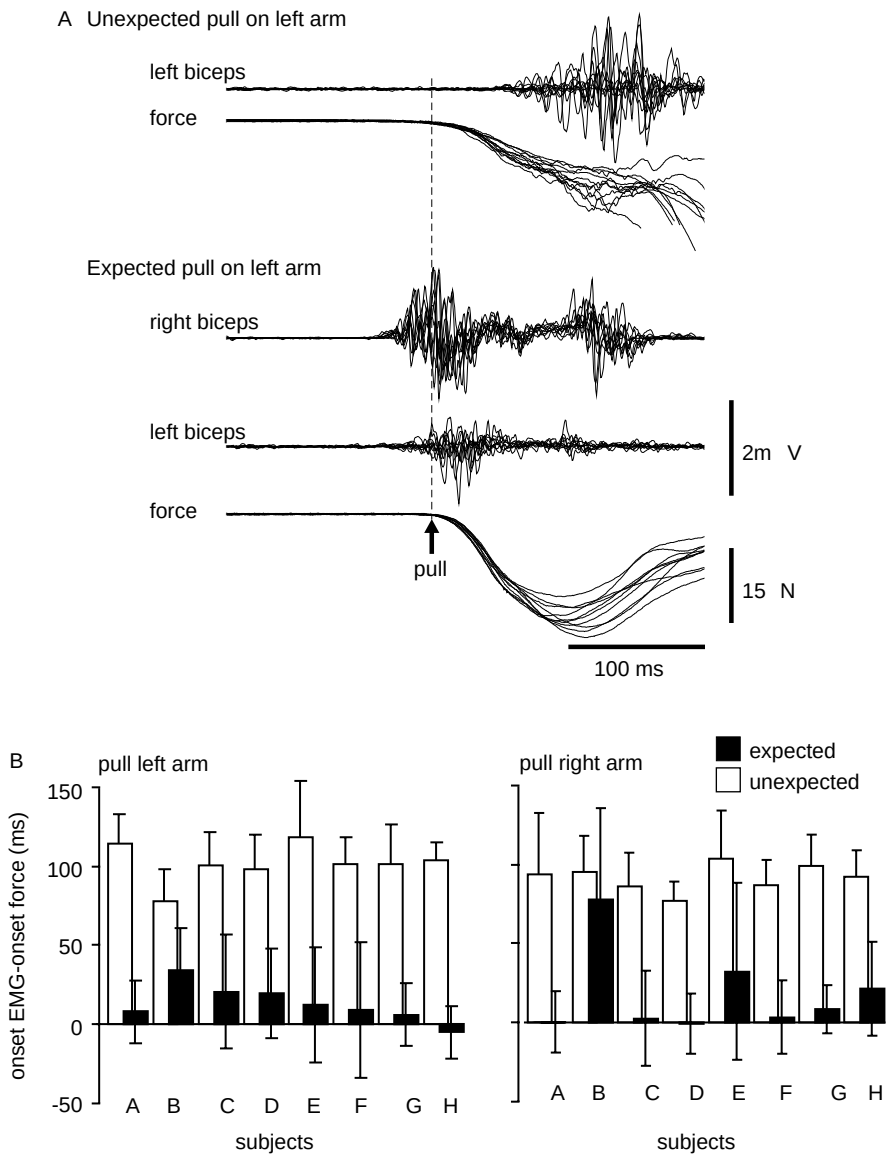

Fig. 2. Responses to an unexpected pull on the arm compared to a pull created by voluntary movement of the other arm. A. Overlaid raw traces from a single subject. The top two sets of traces show EMG recorded from left biceps brachii and the force exerted on the arm by an unexpected pull. The EMG response occurs after the onset of change in force. The lower three sets of traces show EMG from right and left biceps brachii and the tension (force) in the rope joining the arms. In these trials, the subject voluntarily flexed the right elbow in response to a 'go' signal. EMG in the right arm shows this reaction time response. The EMG in the left arm shows the anticipatory postural response which has an onset before the movement of the right arm causes a change in force on the left. B. Mean time from onset of force to onset of EMG response in each arm in each subject. When the arm was pulled unexpectedly (open columns) reflex EMG responses were evoked at an average of 97 ms after the onset of the extension pull on the arm. When the subject made a reaction time response with one arm (filled columns), EMG responses in the other arm occurred at significantly shorter latencies in all but one subject (B).

regardless of whether the muscle was the prime mover in the task or was maintaining a posture. However, stimulation did not significantly change the onset of EMG in the ipsilateral biceps brachii in any condition (see Fig. 3). Table 1 shows the mean latencies of onset of the EMG in each muscle in each condition.

Delay in the onset of EMG was calculated for each subject by comparing the onset of EMG after magnetic stimulation of the cortex to the onset of EMG in control trials which were mixed in the same set. When the right arm was the prime mover, stimulation over the left hemisphere delayed the onset of voluntary EMG in the right arm by 46 ± 28 ms. Stimulation over the right hemisphere delayed the postural response in the left arm by 96 ± 31 ms. When the left biceps brachii was the prime mover, right hemisphere stimulation delayed its onset by 77 ± 34 ms and stimulation over the left hemisphere delayed the postural response in the right arm by 52 ± 41 ms (Fig. 4). Comparison of these delays using a 2-way repeated-measures ANOVA showed that the delay in the onset of EMG in the left biceps brachii was greater than in the right muscle ($p = 0.023$). However, the postural EMG response was not significantly more delayed than the prime mover activity ($p = 0.08$).

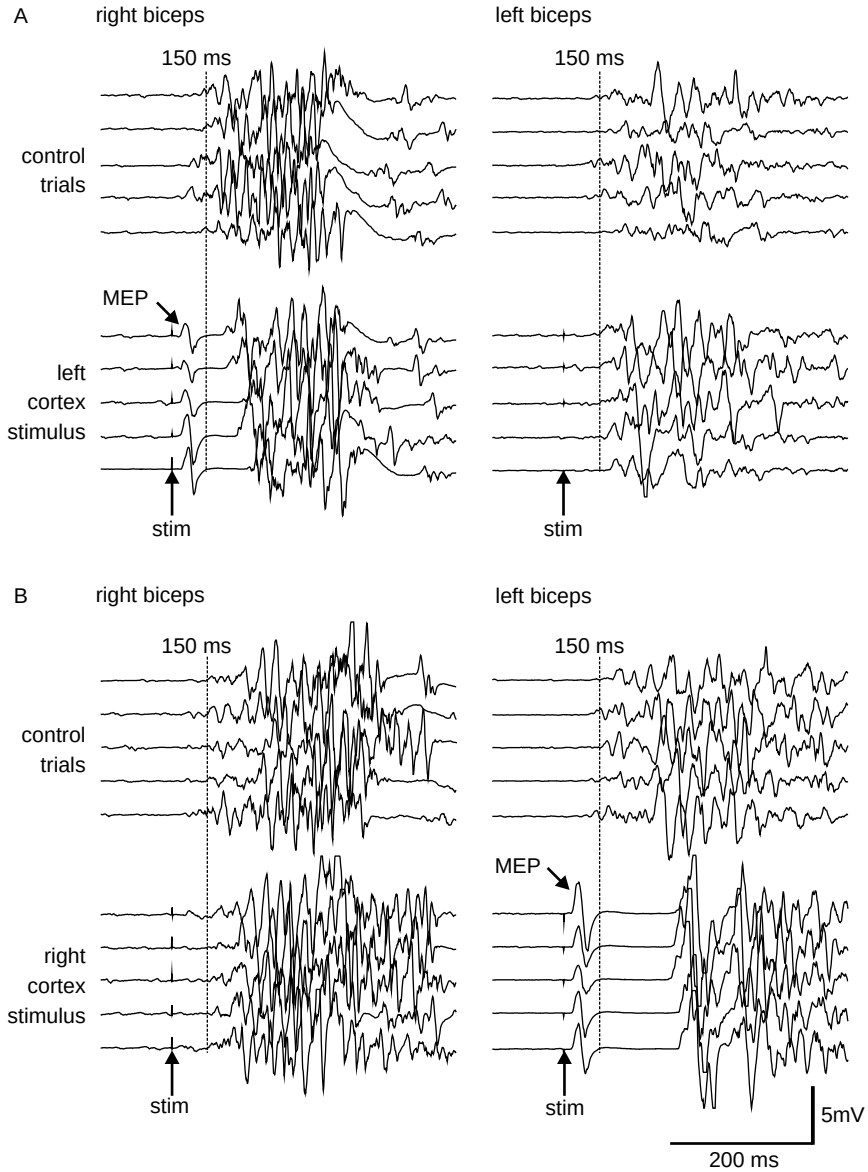

Fig. 3. Transcranial magnetic stimulation during right arm voluntary movement with left arm anticipatory postural responses. Raw EMG traces from right and left biceps brachii from the same trials in a single subject are shown. The right arm contraction was a reaction time response to a visual 'go' signal which occurred at the start of the traces. A. Transcranial magnetic stimulation over the left motor cortex prior to the start of the reaction time response in the right arm. The upper sets of traces show the reaction time response in the right biceps and the anticipatory postural response in the left biceps in control trials with no stimulation. The lower traces show how the responses are affected by stimulation of the left motor cortex at 100 ms after the 'go' signal. A motor evoked potential (MEP) is elicited in the right biceps. The onset of EMG in these traces is delayed compared to that in the control trials. The postural responses evoked in the left biceps in the same trials are shown on the right. The cortical stimulus is ipsilateral to this muscle so that there is no MEP. There is also no change in timing of the onset of the postural response despite the delay of the intended voluntary movement of the other arm. B. Transcranial magnetic stimulation over the right motor cortex. Again, the upper traces show control trials, the right arm was flexed in response to the 'go' signal and the left arm shows a postural response. The lower traces show the effect of stimulation over the right motor cortex. No MEP is seen in the right biceps. The onset of EMG is unaffected by the stimulus. In contrast, the EMG traces recorded from the left biceps brachii show an MEP following the stimulus. The onset of the anticipatory postural response is delayed compared to its onset in the control trials.

3.3. Stimulus intensity matched to produce similar silent periods

In a second study ($n=5$), the intensity of transcranial magnetic stimulation was adjusted for each hemisphere of each subject to cause a silent period of about 180 ms in the EMG of the contralateral biceps brachii during a voluntary contraction. Left motor cortex stimuli ranged from 90 to 95% of stimulator output and right motor cortex stimuli from 85 to 95% stimulator output. In 3 of 5 subjects, the intensity of stimulation of the left cortex was higher than that of the right whereas in one subject, the right was higher than the left and in one subject, intensities were the same. In no subject was the difference in stimulus intensity more than

Table 1
Mean reaction times (ms) for each arm in each condition with a constant stimulation intensity (80% stimulator output)

Matched stimulus intensities	Left arm prime mover		Right arm prime mover	
	Left biceps	Right biceps	Left biceps	Right biceps
Control (no stimulus)	150±18	191±31	183±20	146±16
Left stimulus	150±24	244±39[a]	184±24	198±47[a]
Control	141±17	192±28	193±45	152±30
Right stimulus	234±55[a]	180±21	287±48[a]	157±23
Control	156±35	202±40	192±37	157±29

[a] Significant differences ($p<0.005$) between stimulation trials and their matching controls (repeated measures ANOVA with post hoc Student–Newman–Keuls comparison between means). Mean±SD are shown.

5% stimulator output. Table 2 shows the mean latency of onset for left and right biceps brachii in each condition. The main finding of the first study is confirmed. Transcranial magnetic stimulation over each hemisphere delayed the onset of EMG in the contralateral but not ipsilateral biceps brachii. The delays of the onset of the contralateral biceps brachii EMG were compared. The delays of prime mover activity in the right biceps and the left biceps were similar (67±26 ms and 75±24 ms respectively), as was the delay of the postural response in the right biceps (67±23 ms). However, the delay of the postural response in left biceps was significantly longer (111±11 ms; $p<0.05$, 2-way repeated-measures ANOVA with post hoc Student–Newman–Keuls test).

Transcranial magnetic stimulation altered the performance of the bimanual task both when stimulation was applied over the cortex contralateral to the postural response and when it was applied contralateral to the prime mover. When no stimulation was given, the anticipatory response did not completely compensate for the prime mover action so that the postural arm moved slightly into extension (1–2 cm at the wrist). If the postural response was delayed, then elbow extension of the postural arm was greater. The flexion movement of the arm making the voluntary movement was also greater and was sometimes poorly controlled. In contrast, if the prime mover was delayed there tended to be movement of the postural arm into flexion before the start of the voluntary movement. The changes in arm position were sometimes complicated initially by the twitch evoked by the MEP. This could cause a small deviation in position before the onset of voluntary or postural EMG in either arm.

4. Discussion

The main finding reported here is that a voluntary movement of one arm and its associated anticipatory postural response in the other arm were not both perturbed by magnetic stimulation over the motor cortex of one hemisphere. Each action was affected independently by stimulation over its contralateral motor cortex. Appropriately timed transcranial magnetic stimulation over the motor cortex of one hemisphere delayed the onset of EMG in the contralateral biceps brachii whether the EMG was the action of the prime mover or a postural response. However, stimulation did not affect the onset of EMG in the ipsilateral arm. The finding indicates that the linked responses are not controlled from the motor cortex of one hemisphere.

Delay of a voluntary movement occurs when a magnetic stimulus is given over the contralateral primary motor cortex shortly before the start of the movement (Day et al., 1989; Pascual-Leone et al., 1992; Taylor et al., 1995; Ziemann et al., 1997). It is likely that this delay is another manifestation of the intracortical inhibition which is seen as the silent period in the EMG of an ongoing contraction and which can also be identified during relaxation by testing

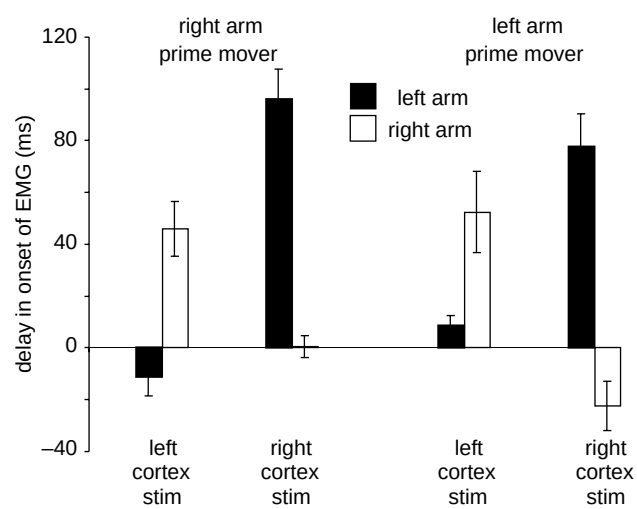

Fig. 4. Delay in onset of EMG caused by transcranial magnetic stimulation. Group data ($n=7$; mean±SEM) from trials in which stimulus intensity was constant (80% stimulator output). The 4 pairs of columns illustrate the delay in EMG in the left biceps (filled column) and right biceps (open columns) in each of the four conditions. For the first pair, subjects made a reaction time movement of the right arm and a postural response occurred on the left. The left motor cortex was stimulated. For the second pair of columns, subjects performed the same task but the right motor cortex was stimulated. For the third pair of columns, subjects made the reaction time response with the left arm and a postural response occurred on the right. The left motor cortex was stimulated. For the fourth pair of columns, subjects performed the left reaction time task with stimulation of the right motor cortex.

Table 2
Mean reaction times in each condition when stimulation intensity was adjusted to match silent periods

Matched silent periods	Left arm prime mover		Right arm prime mover	
	Left biceps	Right biceps	Left biceps	Right biceps
Control (no stimulus)	139±20	154±15	156±30	149±32
Left stimulus	138±18	240±29[a]	153±17	199±38[a]
Control	144±20	173±17	152±19	132±28
Right stimulus	210±30[a]	148±14	254±22[a]	138±19
Control	135±20	168±22	143±22	130±27

[a] Significant differences ($p<0.001$) between stimulation trials and their matching controls (repeated measures ANOVA with post hoc Student–Newman–Keuls comparison between means). Mean±SD are shown.

with a second magnetic stimulus (Chen et al., 1999; Nakamura et al., 1997). The inhibition of corticofugal activity probably results from activation of inhibitory interneurones by the magnetic stimulus, and the consequent actions of GABA on the pyramidal cells (Krnjevic et al., 1966; Roick et al., 1993). In a reaction time movement, inhibition of neurones in the primary motor cortex appears to delay rather than abolish voluntary EMG. This implies that some group of cells, which are either not in the motor cortex or are unaffected by the stimulus for some reason, hold the planned motor response and continue to drive the motor cortex until it responds appropriately (Day et al., 1989). If weak transcranial magnetic stimuli are given before a voluntary reaction, the size of the MEP grows progressively over the interval from around 80 to 100 ms before the onset of voluntary EMG until the EMG starts (Chen et al., 1998; Hashimoto et al., 2004; Pascual-Leone et al., 1992). This increase in the size of the response to stimulation suggests a building excitability in the motor cortex and it is inhibition during this build-up that results in delay of a voluntary response.

The independent delay of associated voluntary and postural responses in the current study has a number of implications. First, the delay of the voluntary action without a delay of the anticipatory postural response indicates that the postural response is not directly driven from the ipsilateral motor cortex via ipsilateral corticospinal neurones. It is also unlikely that the ipsilateral motor cortex indirectly controls the postural response by driving the other motor cortex or any subcortical site, as disruption of the postural response would also be expected in such a case. This is consistent with the retention of contralateral postural responses in patients with callosal transection (Diedrichsen et al., 2005; Viallet et al., 1992). In addition, the occurrence of the postural response well in advance of the delayed voluntary movement confirms that it does not rely on afferent input.

Secondly, the delay of the anticipatory postural response by magnetic stimulation over the contralateral motor cortex suggests that the postural response is driven from the motor cortex in much the same way as a voluntary movement. This is consistent with the severe impairment of the performance and learning of anticipatory responses seen in some hemiparetic patients (Massion et al., 1999; Viallet et al., 1992) but contrasts with the impaired but functional responses seen in others (Bennis et al., 1996). One possible explanation for this inconsistency is that postural responses may be generated via different pathways with recovery after brain injury. Alternatively, it cannot be entirely ruled out that stimulation over the motor cortex could cause a long-lasting inhibition of a remote cortical or subcortical area which generates the postural response. Such an area would require a mechanism, like the motor cortex, to produce the delayed response once the inhibition was ended. However, the delay of the postural response is unlikely to be due solely to effects at a spinal level. Although motoneurones are inhibited for up to 100 ms after activation by the descending volleys evoked by magnetic stimulation (Inghilleri et al., 1993), this inhibition is not long enough to account for the timing of the delayed postural response (mean onset ~165 ms after the stimulus). Furthermore, ongoing background EMG was minimal so that withdrawal of this low level of excitatory drive by inhibition of output from the motor cortex should not prevent excitation of the motoneurones from a different source.

Thirdly, there seems to be no immediate internal feedback between the two responses. The voluntary action is not dependent on the postural response being initiated and the postural response is not dependent on the voluntary action. This contrasts with the monitoring of motor cortical function that apparently occurs with a unilateral action, so that it is delayed rather than inhibited by a transcranial magnetic stimulus (Day et al., 1989). For example, in a unilateral ballistic movement the relative timing of the triphasic EMG bursts (agonist-antagonist-agonist) is maintained. Thus, it seems that even one step prior to motor cortical output, the voluntary action and its associated postural response are controlled independently. The effect of transcranial magnetic stimulation on the two responses is as if their timing is linked by an initial 'go' signal but the responses then develop independently.

The current study does not help identify where such a timing signal might originate nor does it suggest how each of the motor cortices is driven. Massion et al. (1999) also

suggest parallel controls of the intended and anticipatory postural movement. They postulate that the motor cortex responsible for the intended movement may provide timing while the contralateral motor cortex and supplementary motor area generate and modulate the postural response. Transcranial magnetic stimulation delivered close to the onset of EMG would not affect such an arrangement if the timing signal occurred sufficiently early in the motor preparation period. Impaired acquisition of anticipatory responses in Parkinsonian patients (Massion et al., 1999) and in patients with cerebellar degeneration (Diedrichsen et al., 2005) indicate that subcortical structures are important in learning appropriate postural responses. However, Parkinsonian patients had only mild impairment of already learned responses. In patients with cerebellar lesions, previously learned anticipatory responses were also preserved but timing was impaired. This suggests that the cerebellum may be essential for accurate timing of postural responses (Diedrichsen et al., 2005).

The initial set of experiments, in which stimuli of the same intensity were applied to each hemisphere, showed lateralization in the delays of EMG in the right and left arms. EMG on the left tended to be more delayed than that on the right. However, when stimulus intensities were adjusted to give silent periods of matched duration on the two sides, the prime mover was no longer more delayed on the left than the right. Because matching the inhibitory effect of the stimuli on voluntary activity from the motor cortex abolished the difference between the left and right prime movers, it seems likely that the lateralization that was seen initially was due to the effectiveness of the stimulus rather than to a difference in the way in which these motor responses were processed. Although the postural response on the left continued to have a longer delay than on the right, its significance is uncertain. Examination of the timing of postural responses shows that the left arm response was quicker than the right arm response when no stimulus was received whereas the latencies of EMG onset were not greatly different after stimulation (see Table 2). Thus, the differences in delay of EMG in the two arms may be because of the difference in timing of the stimulus relative to the EMG rather than differences between the motor cortices.

This study is consistent with dual control of intended voluntary movements and associated postural responses, with each driven from its contralateral motor cortex. It suggests that after an initial timing signal the intended movement and postural response develop independently.

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